

COLD

REFRIGERATION AND AIR CONDITIONING

Refrigeratio

n

- Branch of engineering sciences that deals with the transfer of heat from a lower temperature region to a higher temperature region, in order to maintain a region at a temperature lower than the surrounding atmosphere
- System used for accomplishing the process of refrigeration is called refrigeration system
- The region which is maintained at a temperature lower than its surroundings is called refrigerated space

Applications of Refrigeration System

- Preservation and transportation of food products
- Preserving life saving drugs, vaccines,...etc
- Used for air conditioning purpose
- Used in medical fields
 - Used for making ice products
- Used for the efficient operation of computers and other work stations



ICE PLANTS



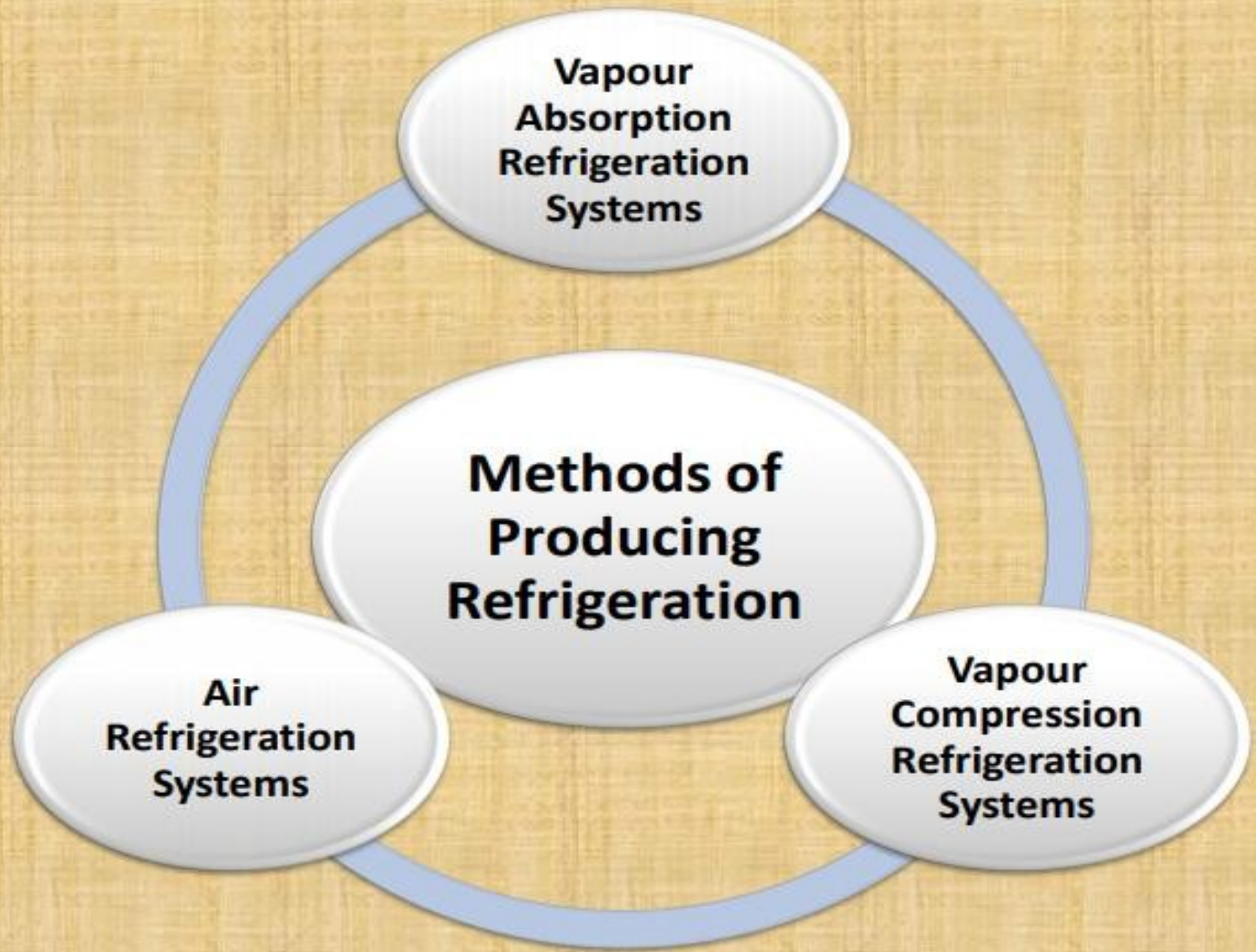
AIR CONDITIONER



REFRIGERATOR

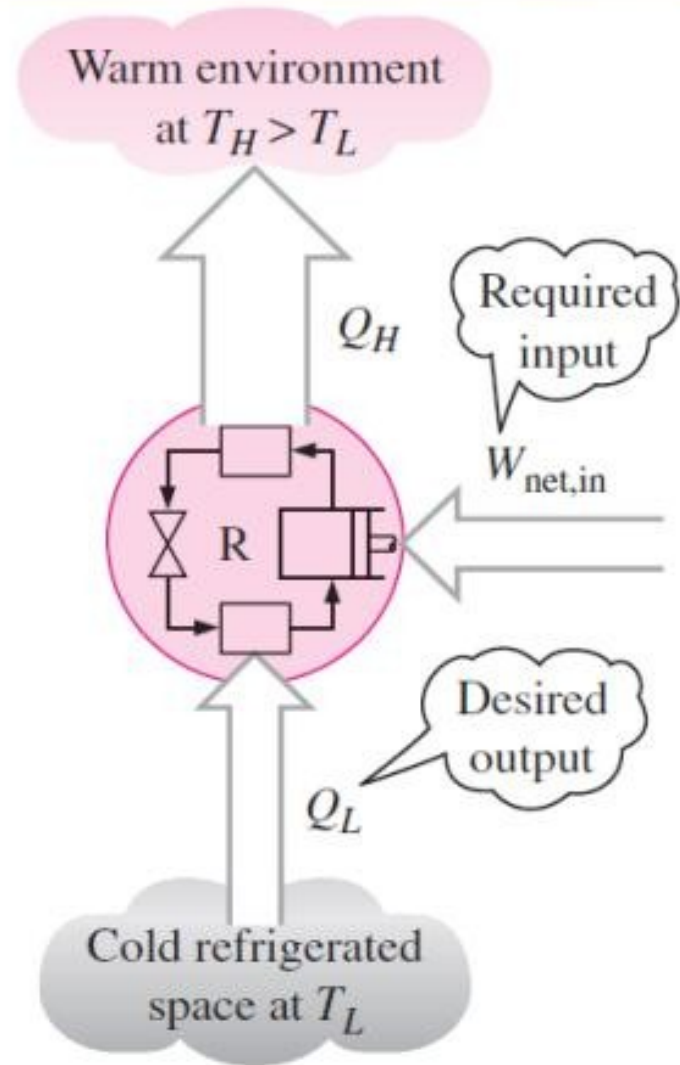


WATER COOLER



Refrigerators

- The transfer of heat from a low-temperature medium to a high-temperature one requires special devices called **refrigerators**
- The objective of a refrigerator is to maintain the refrigerated space at a low temperature by removing heat from it
- Refrigerators, like heat engines, are cyclic devices. The working fluid used in the refrigeration cycle is called a **refrigerant**

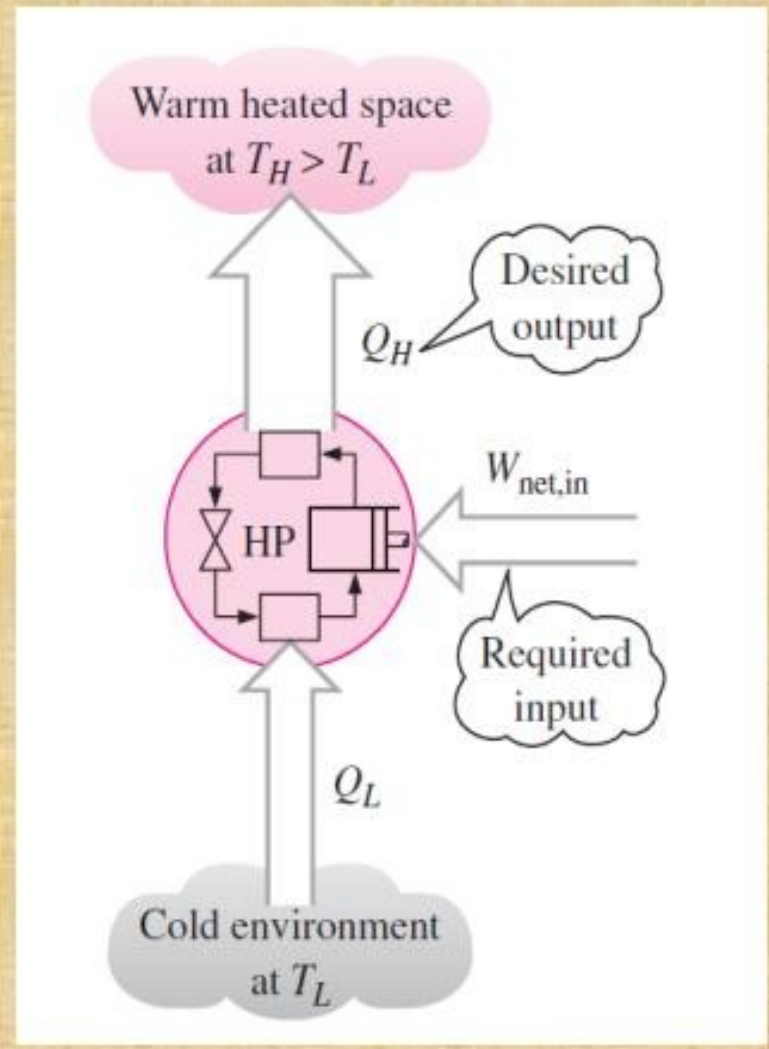


- Q_L is the magnitude of the heat removed from the refrigerated space at temperature T_L , Q_H is the magnitude of the heat rejected to the warm space at temperature T_H , and W_{net} is the net work input to the refrigerator
- The performance of refrigerators are expressed in terms of the **coefficient of performance (COP)**, defined as

$$\text{COP}_R = \frac{\text{Desired output}}{\text{Required input}} = \frac{\text{Cooling effect}}{\text{Work input}} = \frac{Q_L}{W_{\text{net,in}}}$$

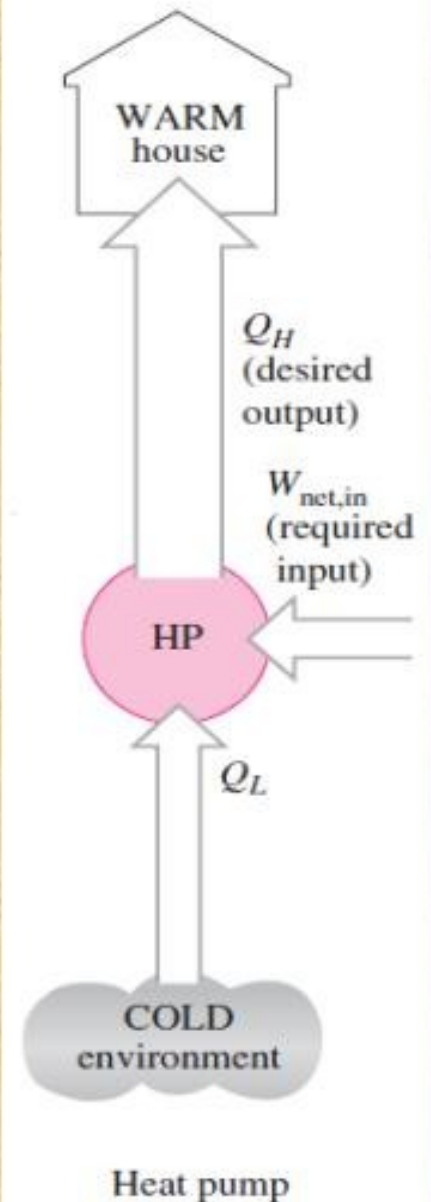
Heat Pumps

- Another device that transfers heat from a low-temperature medium to a high-temperature one is the **heat pump**
- The objective of a heat pump, however, is to maintain a heated space at a high temperature. This is accomplished by absorbing heat from a low-temperature source, such as well water or cold outside air in winter, and supplying this heat to the high-temperature medium such as a house



$$\text{COP}_{\text{HP}} = \frac{\text{Desired output}}{\text{Required input}} = \frac{\text{Heating effect}}{\text{Work input}} = \frac{Q_H}{W_{\text{net,in}}}$$

$$\text{COP}_{\text{HP}} = \text{COP}_{\text{R}} + 1$$



Refrigerating Effect

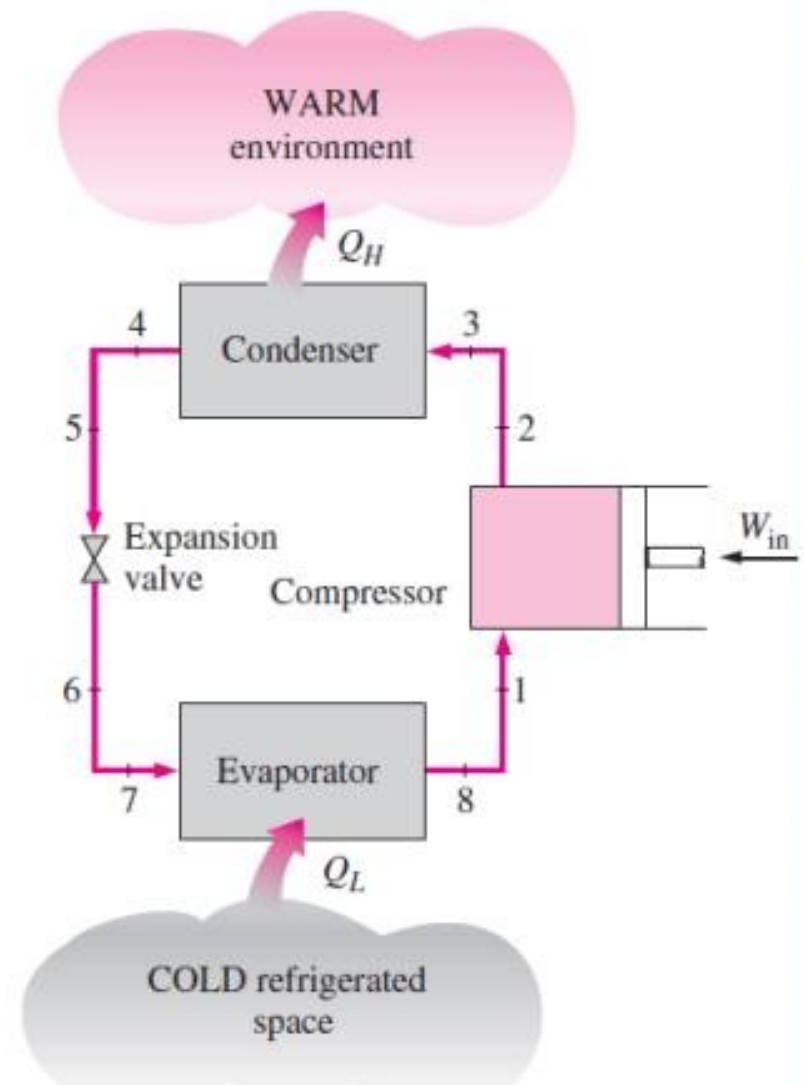
- It is the amount of heat energy removed per unit time from the space to be cooled by the refrigeration process

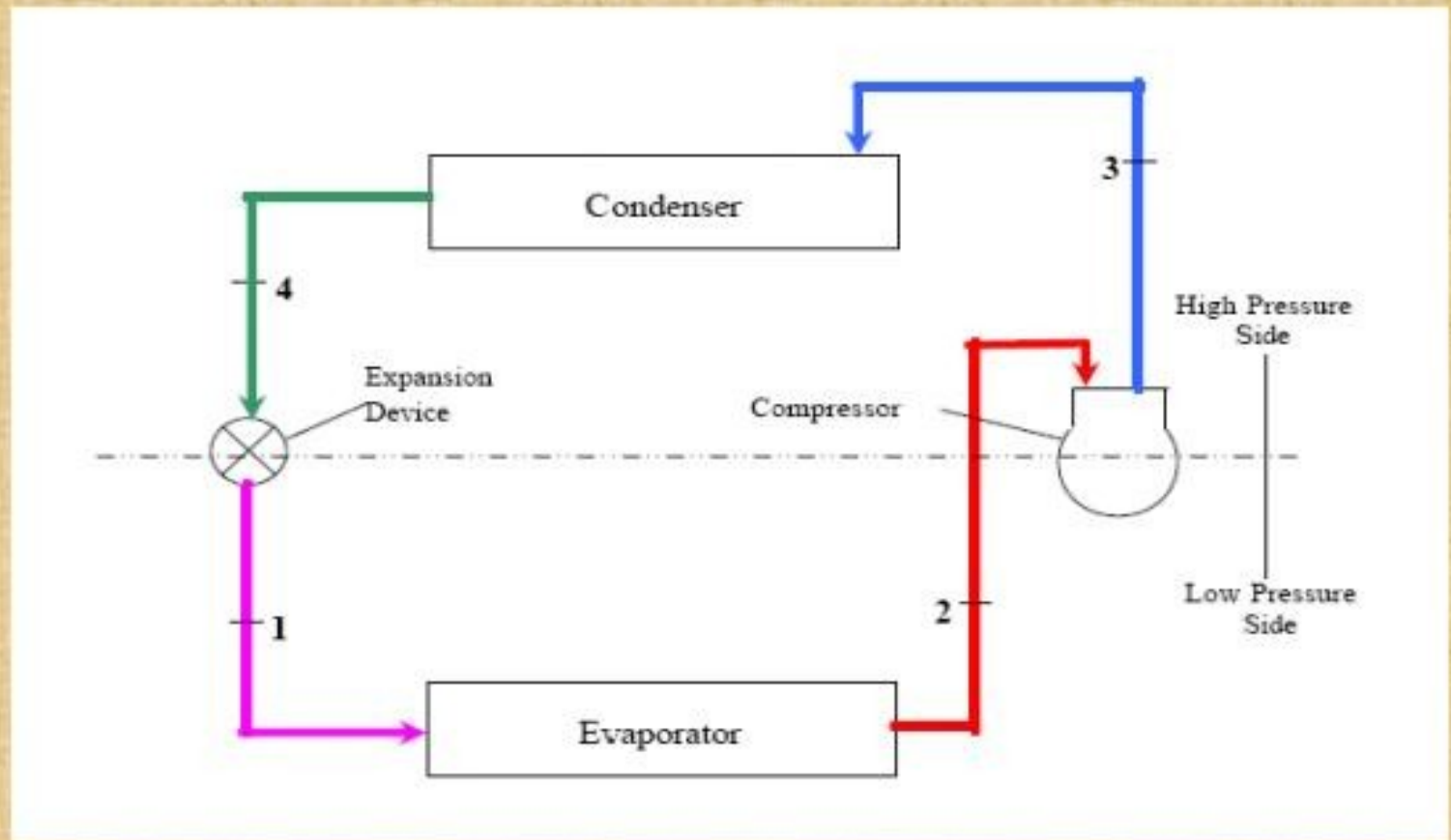
Tons of Refrigeration

- The capacity of a refrigerating machine is generally expressed in tons of refrigeration(TR)
- A ton of refrigeration is defined as the amount of refrigeration effect produced by the uniform melting of one ton (1000kg) of ice from and at 0°C in 24 hours
- One ton of refrigeration(1TR) is equivalent to 210KJ/min or 3.5KW

Vapour Compression Refrigeration System(VCRS)

- In this system, a liquid refrigerant is used which is alternately evaporates and condenses for absorbing heat from the refrigerated space and for rejecting heat to the surroundings
- During the evaporation process, it absorbs heat from and gets converted from liquid to vapour



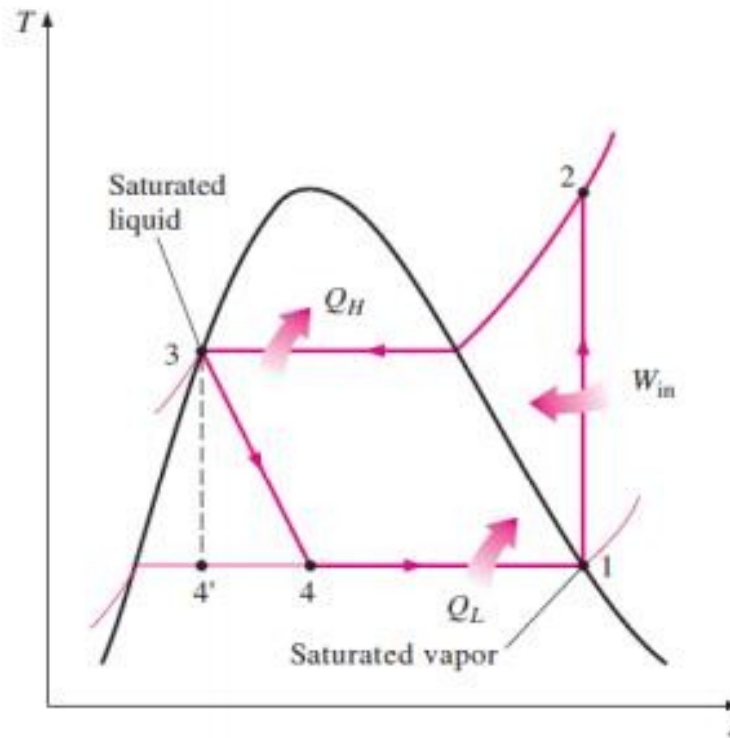
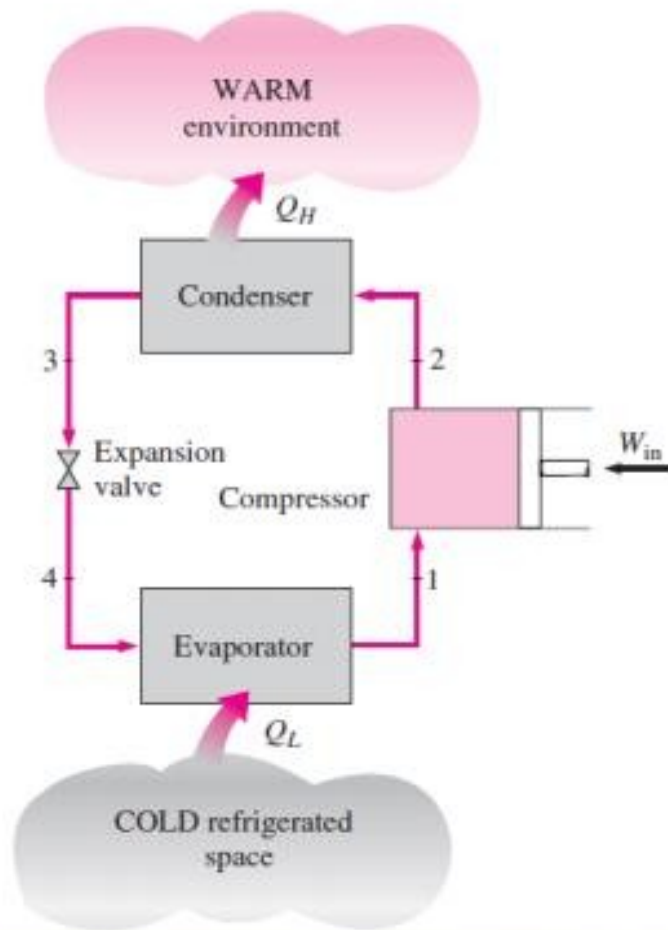


- During the condensing process, it rejects heat and gets condensed from vapour to liquid

Basic components of a VCR

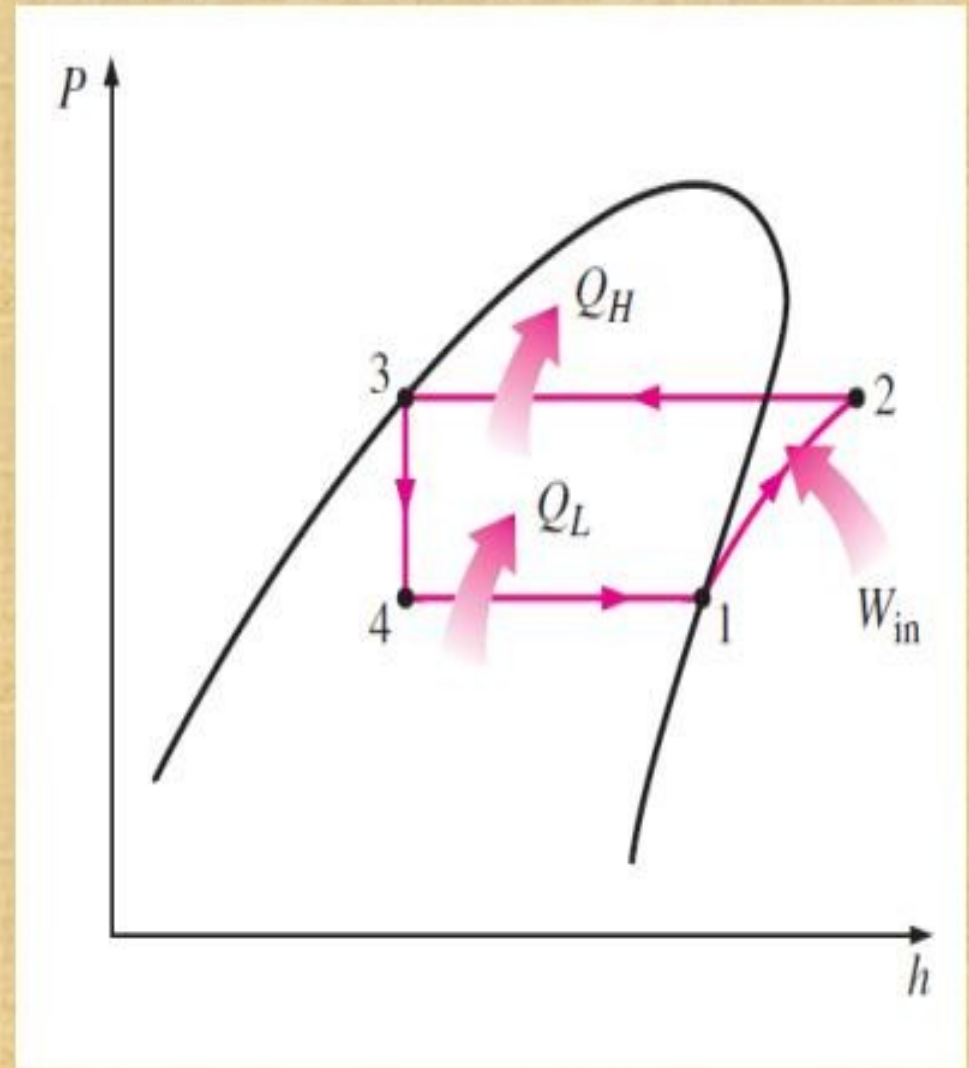
- **Compressor:** Suck the low temperature refrigerant vapour from the evaporator and to compress it to a high pressure and temperature
- **Condenser:** Condense the high pressure and temperature refrigerant vapour from the compressor, by rejecting its latent heat to the cooling water at ambient temperature(or rejects heats to ambient air)
- **Expansion valve(Throttle valve):** Throttle the liquid refrigerant from the condenser. By throttling, the pressure and temperature reduces and the refrigerant becomes partly liquid and partly vapour at the required low temperature. It also regulates the flow of liquid refrigerant to the evaporator
- **Evaporator:** To evaporate the liquid refrigerant by absorbing the latent heat of vapourization from the cold refrigerated space

Working Principle of VCR Systems



It consists of four processes:

- **1-2** Isentropic compression in a compressor
- **2-3** Constant-pressure heat rejection in a condenser
- **3-4** Throttling in an expansion device(constant enthalpy)
- **4-1** Constant-pressure heat absorption in an evaporator



Working

Principle

- In vapour-compression refrigeration cycle, the refrigerant enters the compressor at state 1 as saturated vapour and is compressed isentropically to the condenser pressure. The temperature of the refrigerant increases during this isentropic compression process to well above the temperature of the surrounding medium.
- The refrigerant then enters the condenser as superheated vapour at state 2 and leaves as saturated liquid at state 3 as a result of heat rejection to the surroundings. The temperature of the refrigerant at this state is still above the temperature of the surroundings

- The saturated liquid refrigerant at state 3 is throttled to the evaporator pressure by passing it through an expansion valve or capillary tube. The temperature of the refrigerant drops below the temperature of the refrigerated space during this process
- The refrigerant enters the evaporator at state 4 as a low-quality saturated mixture, and it completely evaporates by absorbing heat from the refrigerated space. The refrigerant leaves the evaporator as saturated vapour and re-enters the compressor, completing the cycle
- The area under the process curve 4-1 represents the heat absorbed by the refrigerant in the evaporator, and the area under the process curve 2-3 represents the heat rejected in the condenser

Psychrometr

- Psychrometry^y is an engineering science that deals with the behaviour of moist air(dry air + water vapour mixture)
- The amount of water vapour in air plays important role in both comfort and industrial air conditioning

Terms used in

Psychrometry

- 1) **Dry air**: Air contains no water vapour
- 2) **Moist air**: Mixture of dry air and water vapour
- 3) **Saturated air**: Air Which contains maximum amount of water vapour which air can hold at a given temperature and pressure
- 4) **Dry bulb temperature**: Temperature of air measured by ordinary thermometer
- 5) **Wet bulb temperature**: Temperature Recorded by a thermometer, when its bulb is covered by a wet cloth
- 6) **Specific humidity**: Ratio of the mass of water vapour to the mass of dry air
- 7) **Relative humidity**: Ratio of mass of water vapour in a given volume of moist air at a given temperature to the mass of water vapour contained in the same volume of moist air at the same temperature, when the air is saturated

Psychrometric

1. **Sensible Heating(C):** Increase the dry bulb temperature of air without change in its specific humidity(moisture content), using heating coil
2. **Sensible Cooling(G):** Decreases dry bulb temperature of air without change in its specific humidity(moisture content), using cooling coil
3. **Humidification(A):** Increase the specific humidity without change in dry bulb temperature, using humidifier
4. **Dehumidification(E):** Decreases the specific humidity without change in dry bulb temperature, using dehumidifier
5. **Heating with Humidification(B):** Increases both dry bulb temperature and specific humidity of air, using high temperature steam
6. **Heating with Dehumidification(D):** Increases the dry bulb temperature and decreases the specific humidity of air, using silica gel or alumina
7. **Cooling with Humidification(H):** Decreases the dry bulb temperature and increases the specific humidity of air, using cold water
8. **Cooling with Dehumidification(F):** Decreases both dry bulb temperature and specific humidity of air, using cooling coil with very low temperature

AIR CONDITIONING



- Air conditioning is the process of controlling and maintaining the internal atmosphere in a confined space. *It involves the control of temperature, humidity, motion of air and purity* of atmosphere in the space of interest
- Air temperature is controlled by cooling or heating the air
- Humidity is controlled by adding or removing water vapour to or from the air
- Air motion is controlled by appropriate air distribution equipment
- Air purity is controlled by filtering and removing undesirable contaminants from the air
- For human comfort, air should have the following properties
 - **Temperature – 22°C to 27°C**
 - **Humidity - 55% to 65% Relative Humidity**
 - **Velocity – 0.3m/s to 0.5m/s**

Applications of Air Conditioning

**Domestic
Applications**

**Industrial
Applications**

**Commercial
Applications**

**Transport
Applications**

**Summer Air
Conditioning**

**Winter Air
Conditioning**

**Air Conditioning
Systems**

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Winter Air Conditioning

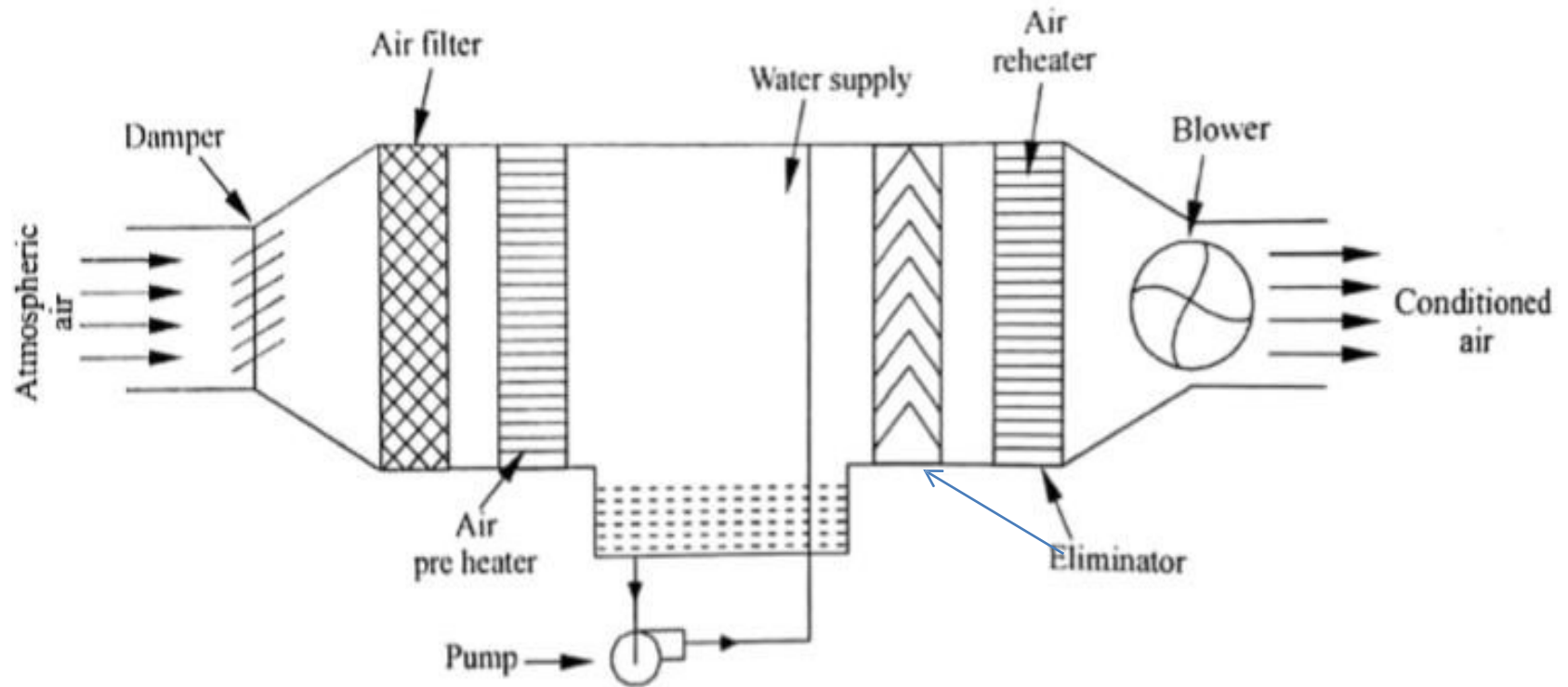
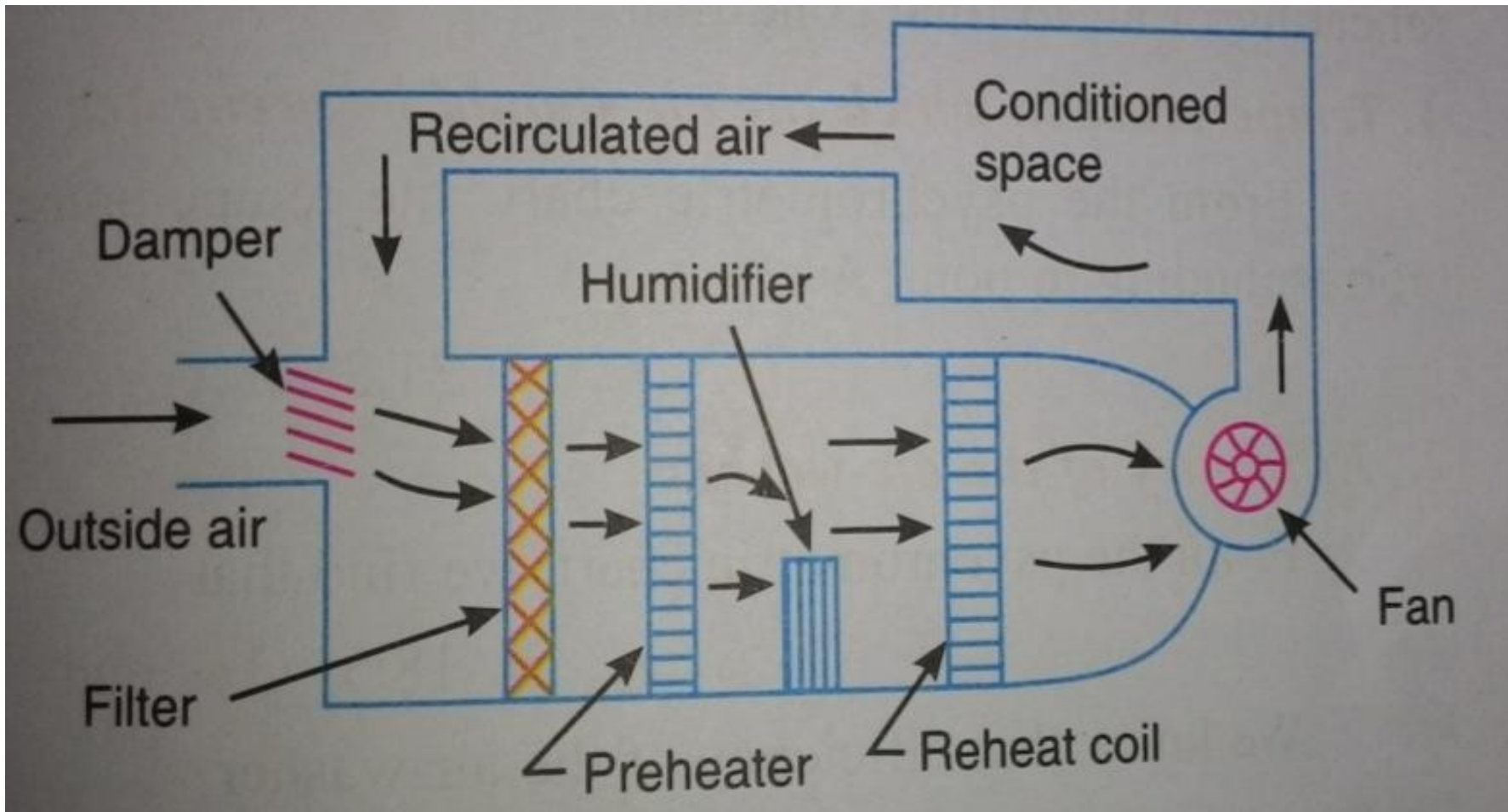


Fig. 5.6. Winter air conditioning

Winter Air Conditioning



Winter Air Conditioning

- *Air is heated and generally humidified*
- *Outside air flows through the damper, and mixes up with recirculated air (obtained from the conditioned space).*
- *The mixed air passes through a filter to remove dust, dirt and other impurities*
- *Air now passes through a preheat coil in order to prevent the possible freezing of water and to control the evaporation of water in the humidifier*
- *The preheated air passes through a reheat coil to bring the air to the designed room temperature*

- The humidification of air can be achieved in several ways, e.g. by bringing the air in contact with a wetted surface, or with droplets of water as in an air washer, by adding water droplets directly to air or by direct addition of dry saturated or superheated steam
- The most common method of humidifying air for these applications is by direct addition of dry steam to air

Summer Air Conditioning

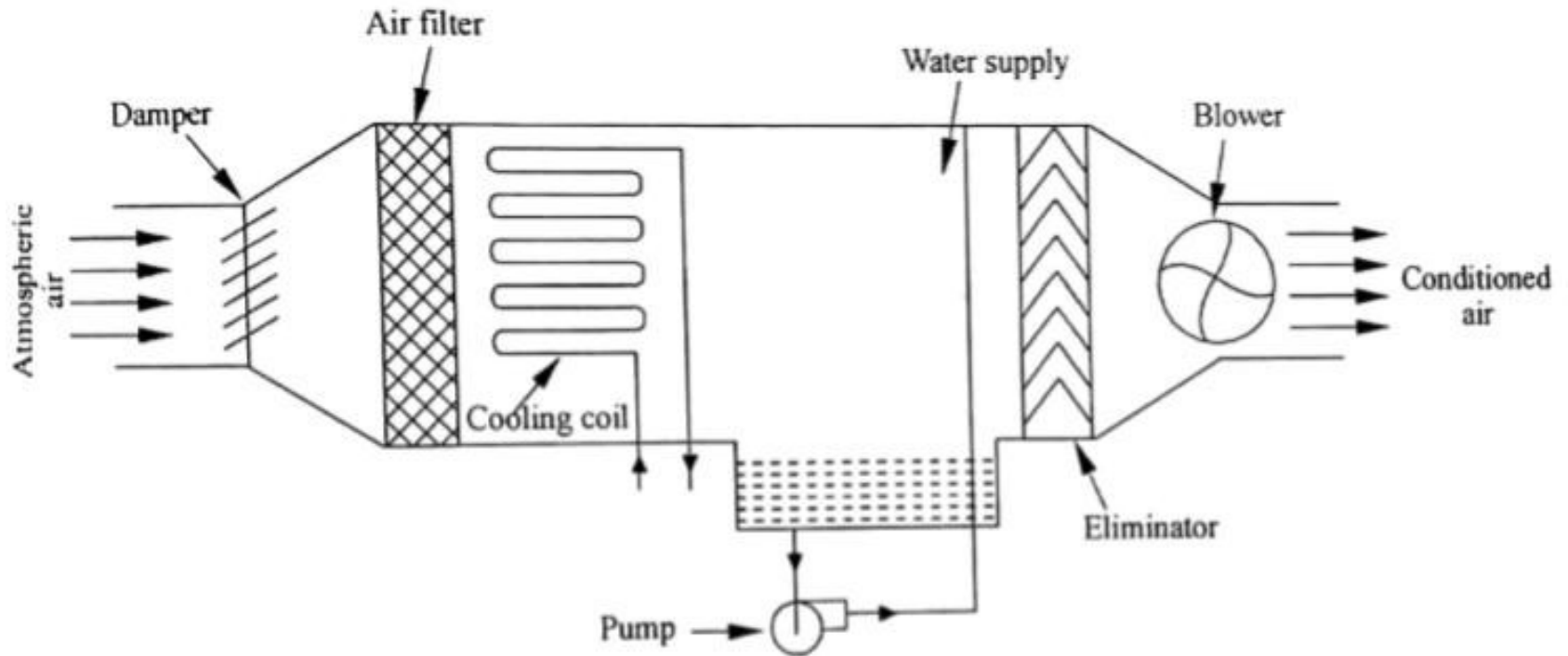
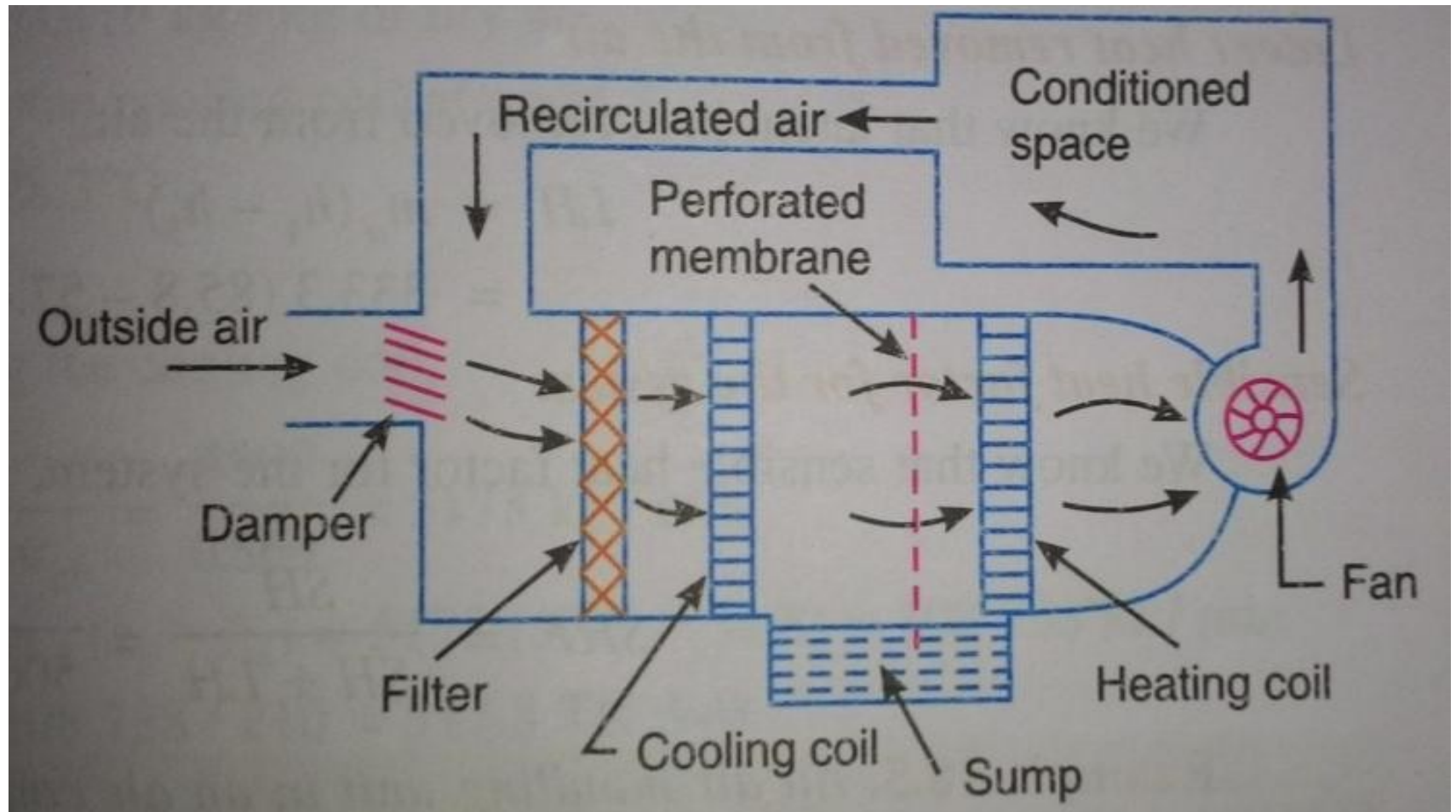


Fig. 5.7. Summer air conditioning

Summer Air Conditioning



- summer air conditioning system requires a refrigeration system that reduces the temperature of the air to temperatures much lower than the surroundings.
- *Air is cooled and generally dehumidified*
- *Outside air flows through the damper, and mixes up with recirculated air(obtained from the conditioned space).*
- *The mixed air passes through a filter to remove dust, dirt and other impurities*
- *Air now passes through a cooling coil. The coil has a temperature much below the required dry bulb temperature of the air in the conditioned space*
- *The cooled air passes through a perforated membrane and loses its moisture in the condensed form which is collected in a sump*
- *Then air is heated by using a heating coil to bring the air to the designed room temperature*

WINTER

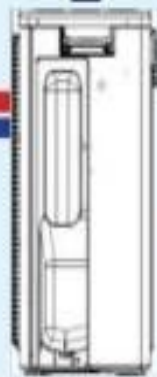


Heated Air

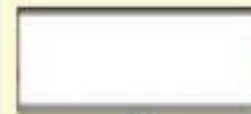
Cool Air

Cooled Air

Outside Air



SUMMER



Cooled Air

Warm Air

Heated Air

Outside Air

